Novia One.

FMP · MODULE 7 · WORKBOOK · V2

OCCUPATIONAL CERTIFICATE · FINANCIAL MARKETS PRACTITIONER

Foreign Exchange Markets workbook answers, V2.

Answers to the Module 7 practical-evidence workbook — all four questions, **90 marks**, written to the marks allocated for each question.

Question 1 · The FX market, exchange rates and money	**30 marks**
Question 2 · Quotations, reciprocals and market players	**25 marks**
Question 3 · Millkorn CC — import payment & exchange control	**10 marks**
Question 4 · Case study — the 2013 forex fixing scandal	**25 marks**
Total	**90 marks**

QUESTION 1 · THE FOREIGN EXCHANGE MARKET [30]

1.1 Define the foreign exchange market and explain its significance in the global financial system. (3)

The foreign exchange market (forex or FX) is the global, over-the-counter market where currencies are bought, sold and exchanged, and it is where the exchange rate for every currency is set. Currencies always trade in pairs, so one currency is always priced against another.

It is significant because it is the largest and most liquid financial market in the world, trading over US$5 trillion a day, 24 hours a day, five days a week. Every cross-border trade, investment and payment has to pass through it, and it is also where businesses and investors hedge their currency risk.

1.2 Differentiate between “free-floating” and “fixed float” currencies and provide examples of each. (3)

A free-floating currency is one whose value is set by the market through supply and demand, without the government fixing the rate. Examples are the US dollar, the British pound and the Japanese yen.

A fixed float is where the government or central bank sets the value of its currency, usually by pegging it to another currency or a standard, so it is not allowed to move freely. Examples are the Saudi riyal and the Panamanian balboa, which are both pegged to the US dollar.

1.3 List and describe the three main types of foreign exchange markets. (3)

- **Spot market** — currencies are exchanged immediately at the current rate, with settlement two business days after the deal. It makes up the largest part of the market.
- **Forward market** — the buyer and seller agree today to exchange currencies at a fixed rate on a set future date. It is traded over the counter and is mainly used for hedging.
- **Futures market** — similar to forwards, with an agreed rate on an agreed future date, but the contracts are standardised and traded on a regulated exchange, which removes counterparty risk.

1.4 Identify and briefly explain four factors that influence currency exchange rates. (3)

Four of the main factors are inflation, interest rates, political stability and the trade balance. Higher inflation weakens a currency because it erodes its buying power. Higher interest rates attract foreign capital looking for a better return, which strengthens the currency. Political stability builds investor confidence and draws in investment, while instability or corruption drives it away. Lastly, a country that exports more than it imports runs a trade surplus, which increases demand for its currency, whereas a trade deficit weakens it.

1.5 What is Purchasing Power Parity (PPP), and how is it used in economic analysis? (3)

Purchasing Power Parity (PPP) is an economic theory that compares currencies using a "basket of goods" approach. The PPP exchange rate is the rate at which the same basket of goods and services would cost the same in both countries. For example, if a shirt costs $10 in the US and the same shirt costs the equivalent of $15 in Germany, then PPP = 15 ÷ 10 = 1.5.

In economic analysis PPP is used to compare economic productivity and standards of living between countries. Some countries also adjust their GDP figures for PPP, so that economies can be compared like for like rather than at market exchange rates.

1.6 Explain the concept of exchange rates and how the foreign exchange market originated. (3)

An exchange rate is the price of one currency in terms of another. It tells you how much of the quote currency you need to buy one unit of the base currency, so USD/ZAR 7.50 means it costs R7.50 to buy one US dollar.

Currency exchange is as old as trade itself, going back to ancient civilisations that traded with metal coins valued by their weight, and the first real forex market started in Amsterdam about 500 years ago. The gold standard (1875) tied currencies to gold, and after World War II the Bretton Woods system pegged currencies to the US dollar, which was backed by gold. When the US ended the dollar's convertibility to gold in 1971, and the gold standard was abolished in 1973, currencies began to float freely, which created the modern forex market.

1.7 Differentiate between fiat currency and representative money. (3)

Fiat currency is not backed by a commodity and has no intrinsic value. Its value comes from the government declaring it legal tender and from people's trust in that government. The rand, and the US dollar since it left the gold standard in 1971, are examples of fiat currency.

Representative money, on the other hand, is backed by a physical commodity such as gold or silver and can be exchanged for that commodity, for example a certificate under the gold standard that could be redeemed for a fixed amount of gold. The key difference is what backs the money: trust in the government versus a commodity.

1.8 Discuss three factors that influence the supply and demand of a currency. (3)

International trade is a big one. When a country exports more than it imports, foreign buyers need its currency to pay for the goods, so demand for the currency rises. Inflation works the other way, because as a currency loses purchasing power it becomes less attractive to hold, so demand falls and the currency weakens. Interest rates also play a role, as higher rates attract foreign investors looking for a better return, which increases demand for the currency and supports its value.

1.9 Explain the difference between currency appreciation and depreciation and their impact on imported goods. (3)

Appreciation is when a currency gains value against another currency, and depreciation is when it loses value.

If the rand appreciates, for example USD/ZAR moves from 7.11 to 6.98, an importer needs fewer rand to buy each dollar, so imported goods become cheaper and this helps bring inflation down. If the rand depreciates, for example from 7.11 to 7.20, more rand is needed for each dollar, so imported goods become more expensive and inflation goes up.

1.10 What are ISO currency codes, and how are they used in forex trading? (3)

ISO currency codes are the standard three-letter codes set by the International Organisation for Standardisation (ISO 4217) to identify each currency, for example USD for the US dollar and ZAR for the South African rand. Usually the first two letters stand for the country and the third for the currency.

In forex trading the codes are combined to quote currency pairs. The first code is the base currency and the second is the quote currency, so USD/ZAR shows how many rand it takes to buy one US dollar. This gives the whole market one standard way of quoting and trading currencies.

QUESTION 2 · QUOTATIONS AND MARKET PLAYERS [25]

2.1 Explain the concept of "bid and offer" in the spot foreign exchange market using the USD/ZAR quotation as an example. Highlight the difference for a bank and a client. Assume the current exchange rate is 7.25/7.26. (5)

In the spot market a bank quotes a two-way price: the bid, which is the rate at which it buys the base currency, and the offer, which is the rate at which it sells it. Quotes are always given from the bank's point of view. In USD/ZAR 7.25/7.26 the US dollar is the base currency and the rand is the quote currency.

For the bank, 7.25 is the rate at which it buys dollars and 7.26 is the rate at which it sells them. The client sees it the other way around: a client selling dollars, such as an exporter, receives R7.25 per dollar, while a client buying dollars, such as an importer paying a foreign supplier, pays R7.26 per dollar. The bank buys low and sells high, and the difference of 0.01 (one cent per dollar) is the spread, which compensates the bank for the risk of holding the position.

2.2 Differentiate between direct and indirect currency quotations with examples. Assume the following rates Rand 7.5050 and Sterling 1.7575. (5)

A direct quotation uses the US dollar as the base currency and shows how much local currency is needed to buy one US dollar. For example, USD/ZAR 7.5050 means one US dollar costs R7.5050. Most currencies, including the rand, are quoted this way.

An indirect quotation works the other way around. The other currency is the base and the quote shows how many US dollars you get for one unit of it. For example, GBP/USD 1.7575 means one British pound buys US$1.7575. By market convention sterling, the euro, the Australian dollar and the New Zealand dollar are quoted this way.

2.3 Calculate the reciprocal bid and offer rates for a given USD/ZAR spot rate of 6.8525/6.8575. Show your calculations. (5)

To find the reciprocal (ZAR/USD) rate, divide 1 by each side of the USD/ZAR quote, then swap the sides so that the smaller number stays on the left as the bid:

New bid = 1 ÷ 6.8575 (the old offer)	**= 0.1458**
New offer = 1 ÷ 6.8525 (the old bid)	**= 0.1459**
ZAR/USD	**0.1458 / 0.1459**

The sides swap because dividing 1 by the bigger number gives the smaller answer, so the old offer becomes the new bid and the old bid becomes the new offer. (The study guide rounds the offer up and shows the answer as 0.1458 / 0.1460.)

2.4 What are the key advantages of using spot exchange rates for businesses? List and explain any three. (5)

- **Simplicity** — the spot rate is simply today's price to exchange one currency for another, so it is easy to understand, with no forward points or contracts to manage.
- **Speed** — deals settle quickly, normally within two business days, which suits businesses that need to pay suppliers promptly.
- **Flexibility and cost** — there is no long-term commitment and no upfront deposit or margin, so a business can deal when the rate suits it and only for the amount it needs.

QUESTION 2 · CONTINUED

2.5 Discuss the different players in the foreign exchange market and their roles. (5)

Commercial banks are the core of the market. They quote two-way prices, deal with each other on the interbank market and handle their clients' foreign payments. Central banks, such as the SARB, manage their country's currency and reserves and can step in to stabilise the exchange rate or carry out monetary policy. Corporations such as importers and exporters use the market for real business needs, buying and selling currency to pay for goods and hedging their currency exposure. Hedge funds, investment managers and sovereign wealth funds trade large amounts to manage their portfolios or to speculate, while brokers and liquidity providers connect buyers and sellers and keep the market liquid. Individual retail traders make up a small part of the market and mainly speculate.

QUESTION 3 · MILLKORN CC — IMPORT PAYMENT & EXCHANGE CONTROL [10]

3.1 Which payment method will Millkorn CC have to consider? (1)

Millkorn CC will have to pay cash in advance (cash with order), meaning it pays the full USD 94,000 before the goods are shipped.

3.2 List at least TWO (2) risks that Millkorn CC is facing, when paying for goods in advance. (2)

- The supplier may never ship the goods, and Millkorn has already paid with no security.
- The goods may arrive incorrect, damaged or not to specification, and because it has already paid, Millkorn has little leverage to get the problem fixed.

3.3 Payment will be made against a "pro-forma invoice", how will Millkorn CC clear the goods at point of arrival (i.e. customs)? (2)

A pro-forma invoice is only a quotation and is not accepted by customs. To clear the goods Millkorn, normally through its clearing agent, needs the supplier's final commercial invoice, the bill of lading and the packing list, and must submit a SARS Customs Declaration and pay any duties and VAT. Because the advance payment is over R100,000, Millkorn must also give its bank a copy of the SARS Customs Declaration (with the MRN) within four months of paying, as proof that the goods were actually imported.

QUESTION 3 · CONTINUED

3.4 Section B.1(D) of the Currency and Exchange Manual for AD covers the Terms of Payment for imports. Explain the difference between Capital Goods and Consumer Goods. Provide at least ONE (1) example of each. (4)

Capital goods are tangible assets, such as property, plant and equipment, that a business uses to produce goods or provide services and that are used over more than one period. Examples are the drying kilns Millkorn is importing, sawmill machinery or a forklift.

Consumer goods are items bought by the end user for their own personal or household use, which are consumed rather than used to produce something else, for example clothing, food or a television. The difference matters under B.1(D), because capital goods may be paid for in advance up to 100% of the ex-factory cost (if the total is no more than R10 million), while other imports are paid in advance against an invoice.

3.5 Considering the goods that Millkorn CC is importing, under which section of the Exchange Control Ruling will it be paid? Motivate your answer. (1)

Section B.1(D)(ii), Capital goods. The kilns are equipment that will be used to produce goods over many years, so they are capital goods, and at USD 94,000 the order is well under R10 million, so up to 100% of the ex-factory cost can be paid in advance.

QUESTION 4 · CASE STUDY — THE FOREX FIXING SCANDAL [25]

4.1 How did the 2013 forex scandal come to light? (2)

It came to light when Bloomberg News reported in June 2013 that currency dealers had been front-running client orders and rigging the WM/Reuters benchmark rates. This led regulators in Asia, Switzerland, the UK and the US to investigate, and suspicious price movements had also been flagged by a whistleblower.

4.2 How were the exchange rates rigged? (2)

The fix was calculated from actual trades done in a 60-second window around 4pm London time. Traders at the big banks colluded in private chatrooms (such as “The Cartel” and “The Mafia”), shared confidential client order information, and then placed large buy or sell orders in that window, known as “banging the close”, to push the benchmark rate in their favour.

4.3 Was there a regulatory failing? Motivate your answer. (2)

Yes. Regulators left the forex market largely unregulated because they believed it was "too big to be manipulated", and they missed early warning signs, such as a 2006 Bank of England meeting at which market manipulation was apparently discussed in front of officials. This light-touch approach effectively invited the banks to game the rules.

4.4 Can such scandals be prevented? (2)

Not completely, but they can be made much harder. Stricter supervision, heavy fines, the longer five-minute fix window and closer desk-by-desk monitoring by managers should stop this kind of cheating for a while. However, there is no room for complacency, because corporate memories are short and the rewards for beating the market are big.

4.5 By how many minutes has the daily 4pm fix been extended to? (1)

The fix window has been extended from one minute to five minutes.

4.6 What was the estimated amount that was lost due to the manipulation of the forex market? (1)

An estimated $11.5 billion (£7.5 billion) a year for Britain's 20.7 million pension holders alone, as at December 2014.

4.7 Which FIVE banks were fined by the Financial Conduct Authority (FCA)? (5)

- Citibank ($358 million)
- HSBC ($343 million)
- JPMorgan ($352 million)
- Royal Bank of Scotland, RBS ($344 million)
- UBS ($371 million)

4.8 What was the finding about managing risk? (2)

The FCA found that between 1 January 2008 and 15 October 2013 the five banks failed to manage the risks around client confidentiality, conflicts of interest and trading conduct. This allowed their traders to use confidential client order information to collude with other banks and manipulate G10 exchange rates for their own profit.

4.9 What is your understanding of the G10 currencies? List them. (5)

The G10 currencies are the ten most heavily traded and most liquid currencies in the world. They are issued by large, developed economies with stable governments and freely floating exchange rates, and they make up most of global forex trading. They are:

- US dollar (USD)
- Euro (EUR)
- Japanese yen (JPY)
- British pound (GBP)
- Swiss franc (CHF)
- Canadian dollar (CAD)
- Australian dollar (AUD)
- New Zealand dollar (NZD)
- Swedish krona (SEK)
- Norwegian krone (NOK)

4.10 Is the Forex Market regulated in South Africa? Motivate your answer. (2)

Yes. The Financial Sector Conduct Authority (FSCA) regulates forex trading and the providers who offer it, and the SARB controls the flow of currency through the exchange control regulations. Only Authorised Dealers (ADs) and Authorised Dealers with Limited Authority (ADLAs) appointed by the SARB may buy and sell foreign currency, and South African residents may not speculate against the rand.

4.11 Who authorises dealing in foreign exchange transactions in South Africa? (1)

The South African Reserve Bank (SARB), which appoints the banks as Authorised Dealers to deal in foreign exchange.